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Optical Measurement System: Intensity & Wavelength Modulation

ICPE- 356 Advanced Measurement Systems

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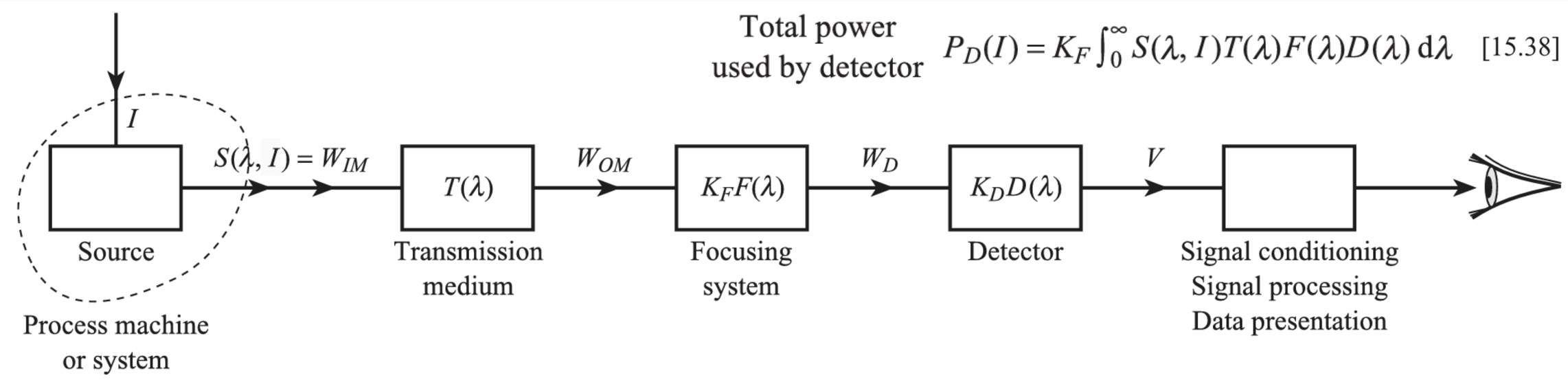
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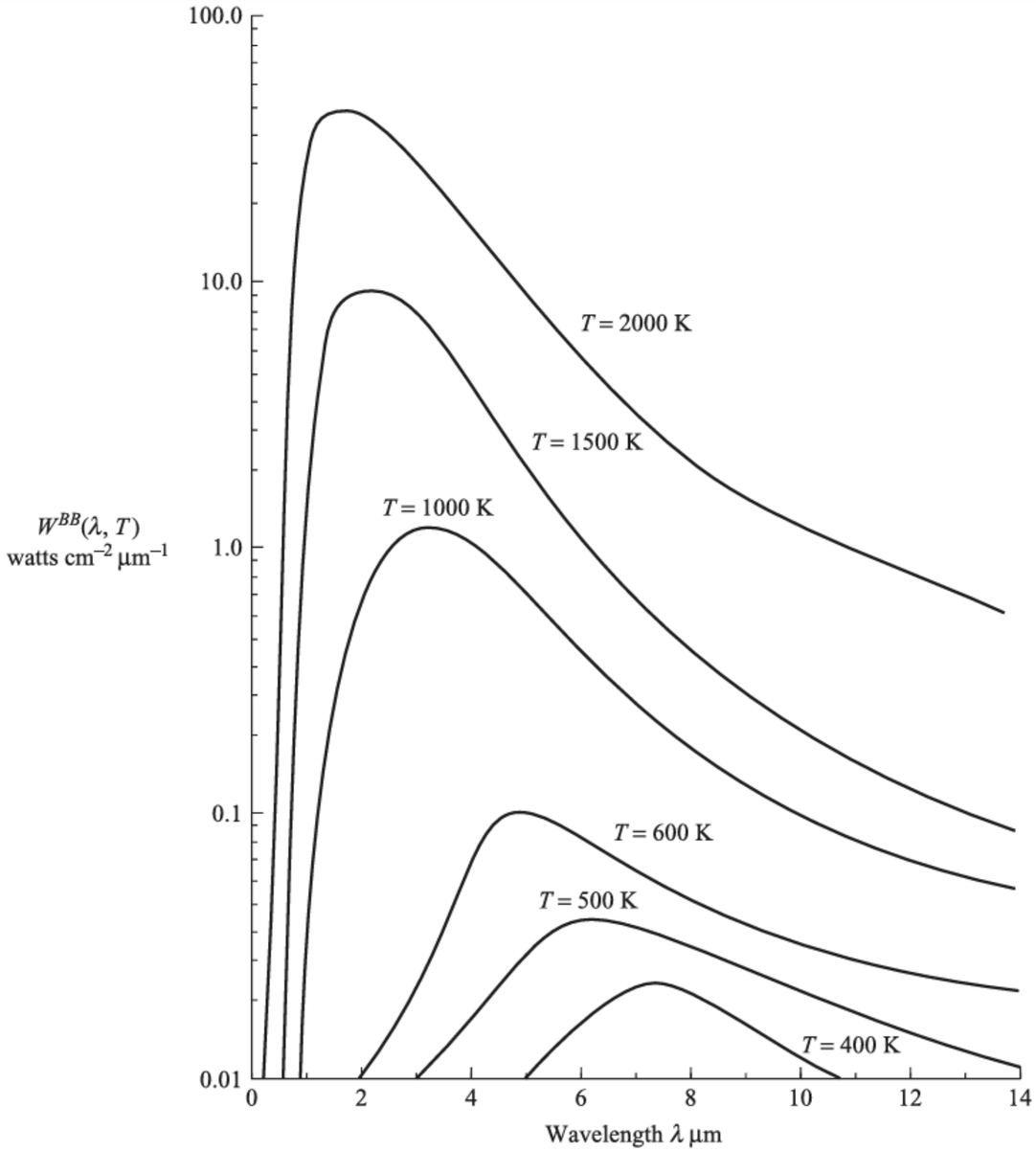
Variable source, fixed transmission medium



The function **S(λ)** describes how the amount of radiant power emitted by the source varies with wavelength λ. **T(λ)** describes how the efficiency of the transmission medium varies with λ. The detector converts the incoming radiant power into an electrical signal. The sensitivity or responsivity K_D is the change in detector output (ohms or volts) for 1 watt change in incident power. **D(λ)** describes how the detector responds to different wavelengths λ.. The values of all three functions must be reasonably large over the wavelengths of interest, otherwise these wavelengths will not be transmitted.

The system shown has a variable source in that the source power S varies according to the value of the measured variable I. In this system the transmission medium is usually the atmosphere with fixed characteristics T(λ). A focusing system is usually necessary to couple the source to the detector. The constant K_F specifies the efficiency of this coupling and the function **F(λ)** the wavelength characteristics of the focusing system. This means they can be used in situations where it is impossible to bring conventional sensors, into physical contact with the process.

Modulation of intensity by source



Any body at a temperature above 0 K emits radiation. The ideal emitter is termed a **black body**: from Planck's Law the spectral exitance of radiation emitted by a black body at temperature T K is:

$$W^{BB}(\lambda, T) = \frac{C_1}{\lambda^5 \left[\exp\left(\frac{C_2}{\lambda T} \right) - 1 \right]} \qquad [15.6]$$

where $W^{BB}(\lambda, T)$ is the power emitted per unit wavelength at λ.

λ = wavelength in µm

$C_1 = 37\,413$ W µm^4 cm^{-2}

$C_2 = 14\,388$ µm K

Figure 15.4 is an approximate plot of $W^{BB}(\lambda, T)$ versus λ, for various values of T.

Summarising, a hot body source is characterised by a broadband, continuous spectral exitance $S(\lambda, T)$ given by:

$$S(\lambda, T) = \frac{1}{\pi} \varepsilon(\lambda, T) W^{BB}(\lambda, T) \quad \text{W µm}^{-1}\,\text{cm}^{-2}\,\text{sr}^{-1} \qquad [15.10]$$

Since the power emitted by a hot body source depends critically on source temperature, systems of this type are commonly used for remote temperature measurement. There are two basic types of system: **broadband** and **narrowband**.

Broadband

These systems use all wavelengths present in the incoming radiation and are often referred to as total radiation pyrometers. A thermal detector is therefore used, and from eqns [15.38], [15.10] and [15.6] the total power used by the detector is:

$$P_D = \frac{C_1 K_F}{\pi} \int_0^\infty \frac{\varepsilon(\lambda, T)\, T(\lambda)\, F(\lambda)\, \mathrm{d}\lambda}{\lambda^5 \left[\exp\left(\frac{C_2}{\lambda T} \right) - 1 \right]} \qquad [15.51]$$

The integral can be evaluated numerically if $\varepsilon(\lambda, T)$, $T(\lambda)$ and $F(\lambda)$ are known. However, if we assume that $\varepsilon(\lambda, T)$, $T(\lambda)$ and $F(\lambda)$ have constant values ε, T_M and F respectively for all λ, T, then using eqn [15.7]

$$P_D = \frac{1}{\pi} \varepsilon T_M K_F F \int_0^\infty \frac{C_1 \mathrm{d}\lambda}{\lambda^5 \left[\exp\left(\frac{C_2}{\lambda T} \right) - 1 \right]} = \frac{1}{\pi} \sigma \varepsilon T_M K_F F T^4$$

The corresponding detector output signal is:

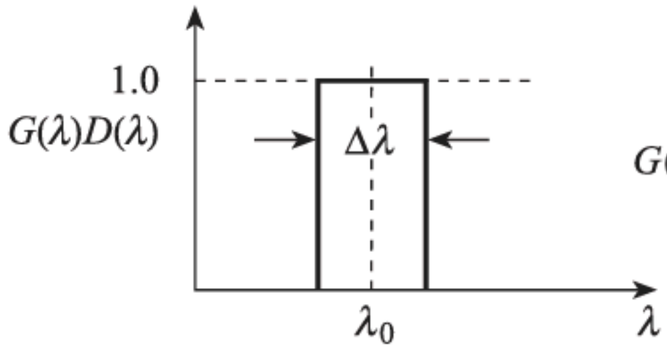
$$V_T = K_D P_D = \frac{K_D}{\pi} \sigma \varepsilon T_M K_F F T^4 \qquad [15.52]$$

The above assumptions represent a considerable oversimplification, and even if they are justified there is still the problem of uncertainties in the values of ε, T_M , etc. For these reasons the system should be calibrated experimentally using a standard source of known emissivity. If the system is then used with a process source of different emissivity then a correction must be made.

Narrowband

These systems use photon detectors so that only a narrow band of wavelengths is used. In some systems a narrow pass-band radiation filter G(λ) is used to further restrict the range of wavelengths incident on the detector. For these systems, the wavelength response G(λ)D(λ) of the combined filter and detector can be approximated by the ideal band-pass filter characteristics, thus the expression for total power used is:

$$P_D = \frac{1}{\pi} C_1 K_F \int_0^{\infty} \frac{\varepsilon(\lambda, T)\, T(\lambda)\, F(\lambda)\, G(\lambda)\, D(\lambda)\, \mathrm{d}\lambda}{\lambda^5 \left[\exp\left(\frac{C_2}{\lambda T} \right) - 1 \right]}$$



(a)

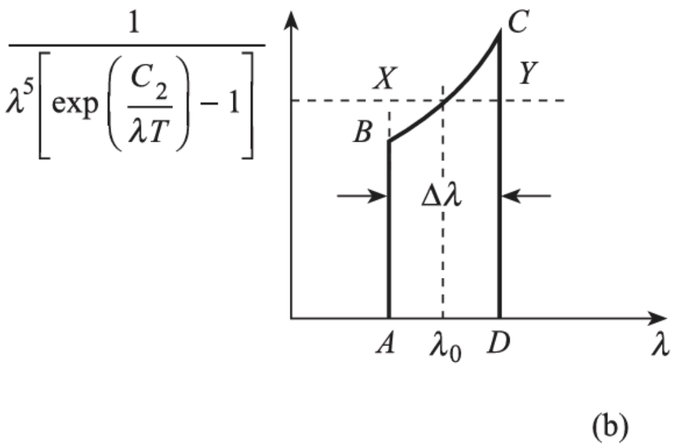
$$G(\lambda)D(\lambda) = \begin{cases} 1.0, & \lambda_0 - \frac{\Delta\lambda}{2} < \lambda < \lambda_0 + \frac{\Delta\lambda}{2} \\ 0, & \lambda > \lambda_0 + \frac{\Delta\lambda}{2} \\ 0, & \lambda < \lambda_0 - \frac{\Delta\lambda}{2} \end{cases}$$

which reduces to:

$$P_D = \frac{1}{\pi} C_1 K_F \int_{\lambda_0 - \Delta\lambda/2}^{\lambda_0 + \Delta\lambda/2} \frac{\varepsilon(\lambda, T)\, T(\lambda)\, F(\lambda)\, \mathrm{d}\lambda}{\lambda^5 \left[\exp\left(\frac{C_2}{\lambda T} \right) - 1 \right]}$$

The assumptions $\varepsilon(\lambda, T) = \varepsilon$, $T(\lambda) = T_M$ and $F(\lambda) = F$ are far more justified for a narrow band of wavelengths and give:

$$P_D = \frac{1}{\pi} C_1 \varepsilon T_M K_F F \int_{\lambda_0 - \Delta\lambda/2}^{\lambda_0 + \Delta\lambda/2} \frac{\mathrm{d}\lambda}{\lambda^5 \left[\exp\left(\frac{C_2}{\lambda T} \right) - 1 \right]} \qquad [15.55]$$



$$\int_{\lambda_0-\Delta\lambda/2}^{\lambda_0+\Delta\lambda/2} \frac{d\lambda}{\lambda^5\left[\exp\left(\frac{C_2}{\lambda T}\right)-1\right]}$$

$$= \text{area } ABCD \approx \text{area } AXYD$$

$$= AX \cdot XY$$

$$= \frac{\Delta\lambda}{\lambda_0^5\left[\exp\left(\frac{C_2}{\lambda_0 T}\right)-1\right]}$$

The integral is evaluated approximately in Figure 15.16(b) to give:

$$P_D = \frac{1}{\pi} C_1 \varepsilon T_M K_F F \frac{\Delta\lambda}{\lambda_0^5\left[\exp\left(\frac{C_2}{\lambda_0 T}\right)-1\right]}$$

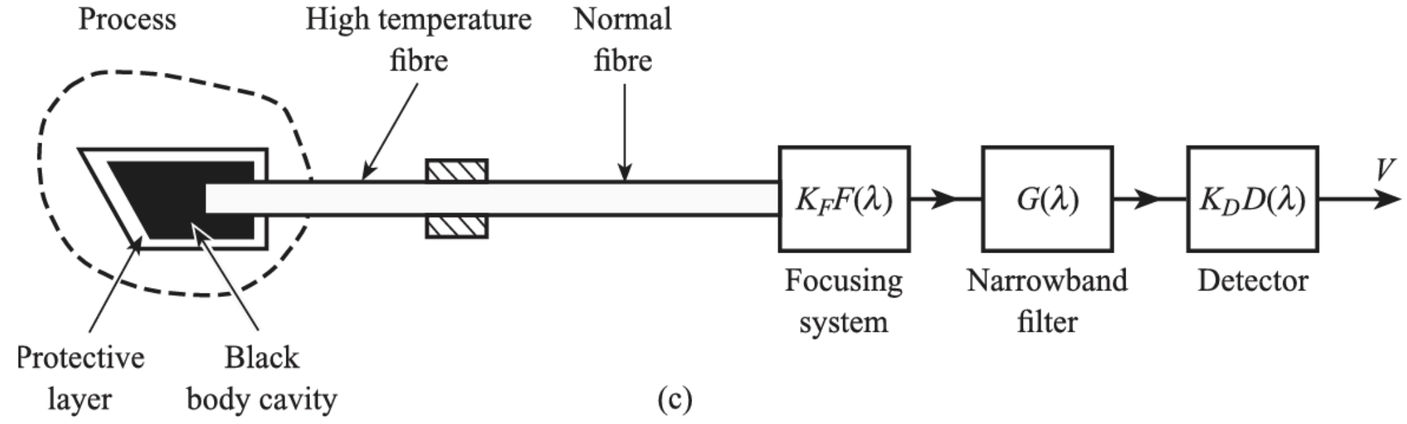
$$\approx \frac{1}{\pi} C_1 \varepsilon T_M K_F F \cdot \frac{\Delta\lambda}{\lambda_0^5} \exp\left(-\frac{C_2}{\lambda_0 T}\right)$$

The approximation is valid since $\exp(C_2/\lambda_0 T) \gg 1$; for example if $\lambda_0 = 1\ \mu m$, $T = 10^3$ K, $(C_2/\lambda_0 T) = 14.4$, $\exp(C_2/\lambda_0 T) = 1.8 \times 10^6$. The detector output signal is thus:

$$V(T) = K_D P_D$$

$$= \frac{1}{\pi} C_1 \varepsilon T_M K_F F K_D \frac{\Delta\lambda}{\lambda_0^5} \exp\left(-\frac{C_2}{\lambda_0 T}\right) \qquad [15.57]$$

The narrow band of wavelengths must be chosen to coincide with a 'window' in the transmission characteristics so that T_M is reasonably large. Again the system should be calibrated experimentally using a standard source of known emissivity and a correction applied if it is subsequently used with a process source of different emissivity.



The above problems of variations of source emissivity and atmospheric transmission characteristics may be solved using the system shown. Here a black body cavity is inserted in the process and radiation is transmitted to the radiation receiver using an optical fibre system rather than the atmosphere. The fibre system consists of a short run of special fibre, which can withstand the high process temperatures, connected to a much longer length of normal fibre.

Modulation Of Intensity By Transmission Medium

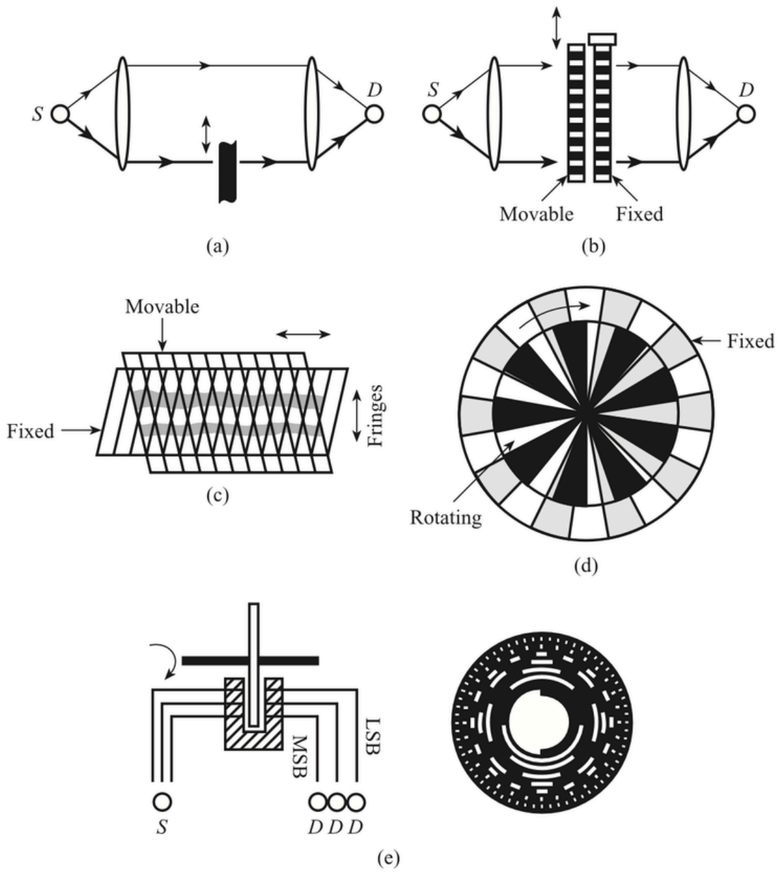
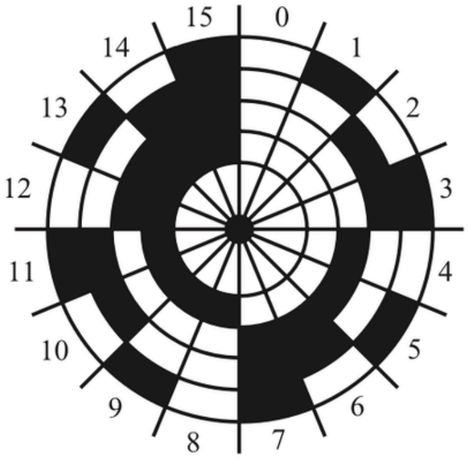


Figure 15.17 Intensity modulation using shutters and gratings:
(a) Shutter modulator
(b) Grating modulator
(c) Linear displacement moiré fringe gratings
(d) Angular displacement moiré fringe gratings
(e) Digital encoder disc
(f) Binary coded disc
(g) Gray coded disc.

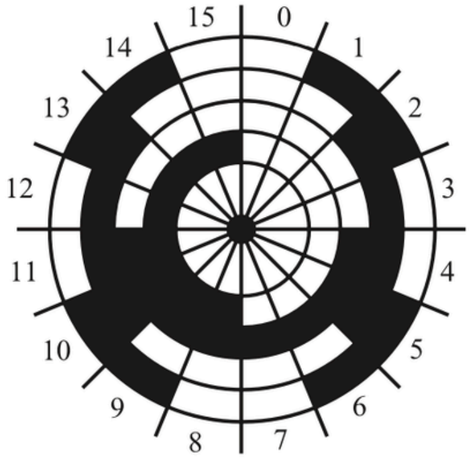
These are systems of the type shown in Figure 15.1, where the measured variable alters the characteristics of the transmission medium and so modulates the intensity of the radiation reaching the detector. There are several ways in which this modulation can be achieved.

The simplest type of shutter displacement sensor is shown in Figure 15.17(a). Figures 15.17(b), (c) and (d) show moiré fringe gratings for the digital measurement of linear and angular position. In the linear type there are two finely ruled gratings

of the same pitch P, slightly angled to each other. Relative movement of the gratings in a horizontal direction causes a series of dark fringes to move vertically across the sensor, giving an approximately sinusoidal variation in light intensity and detector output signal. In the angular case, one grating has N lines, the other $N + 1$ lines (N is typically between 50 and 1200); for every complete revolution of the gratings there are N complete revolutions of the fringe pattern and N cycles of the detector output signal. Thus in both cases the exact displacement can only be established by counting the number of detector output cycles; moiré gratings are therefore often referred to as **incremental encoders**.



(f)



(g)

Figures 15.17(e), (f) and (g) show coded disc systems for the digital measurement of angular position. Here the disc is marked out in transparent and opaque segments according to the required digital code. **Gray code** (Figure 15.17(g)) is a cyclic code in which only one digit can change between two consecutive sectors; this is in contrast to binary code (Figure 15.17(f)), where several digits may change between consecutive sectors. If the detectors are misaligned slightly, then with a binary coded disc false codes can occur during the transition from one sector to the next. This cannot occur with Gray code; Figure 15.17(e) shows a typical eight-digit Gray coded disc. The output digital code uniquely specifies the angular position of the disc, and coded discs are often referred to as **absolute encoders**.

Intensity modulation using relative displacement of fibres

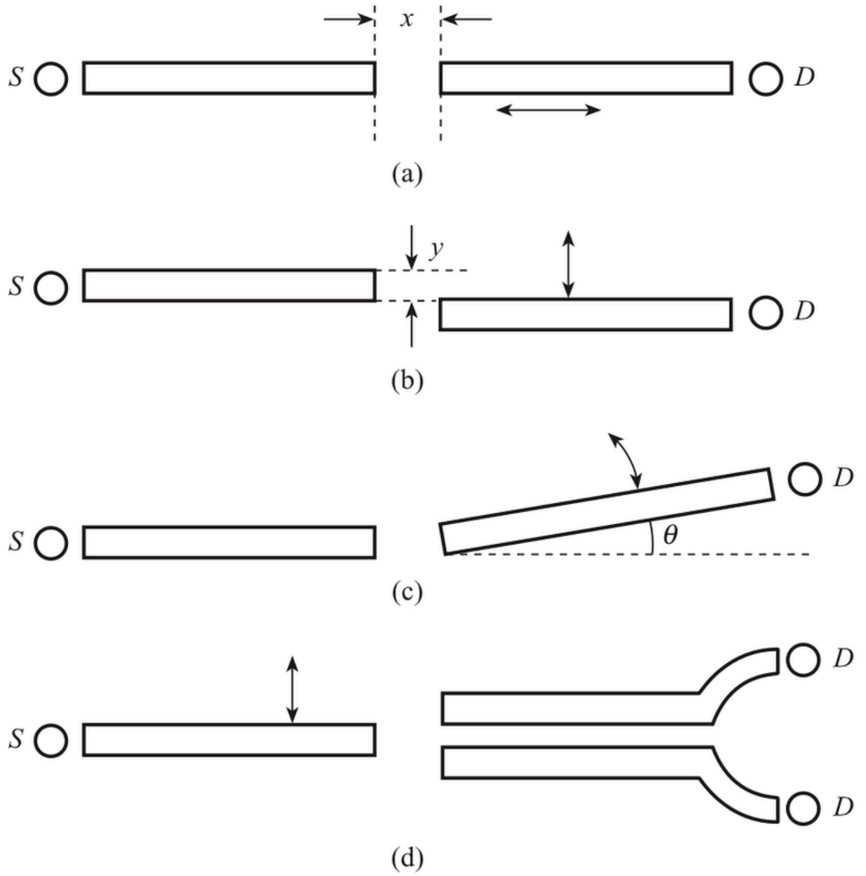


Figure 15.18 Intensity modulation using relative displacement of fibres:
(a) Longitudinal
(b) Transverse
(c) Angular
(d) Differential.

Figure 15.18 shows four possible arrangements for the measurement of small linear or angular displacements. Here the intensity of the radiation incident on the detector is modulated according to the relative displacement of two optical fibres. The underlying principles have already been discussed in Section 15.4.2. For example in Figure 15.18(a), as x is decreased the solid angle ω of the cone of rays into the second fibre increases, causing the power incident on the detector to increase. This effect continues until the cone of rays into the second fibre corresponds to the cone of emergence of the first fibre. The arrangements shown in Figures 15.18(a), (b) and (c) are affected by fibre misalignment, vibration and source variations; the differential system shown in Figure 15.18(d) should be unaffected by source variations.

(b) Transverse displacement

As the second fibre is moved up or down by a small amount y, the overlap between the emission cone of the first fibre and the acceptance area of the second fibre changes. When the axes are closer, more light is captured; as they separate, less is coupled, so the detected power falls off approximately linearly with small y.

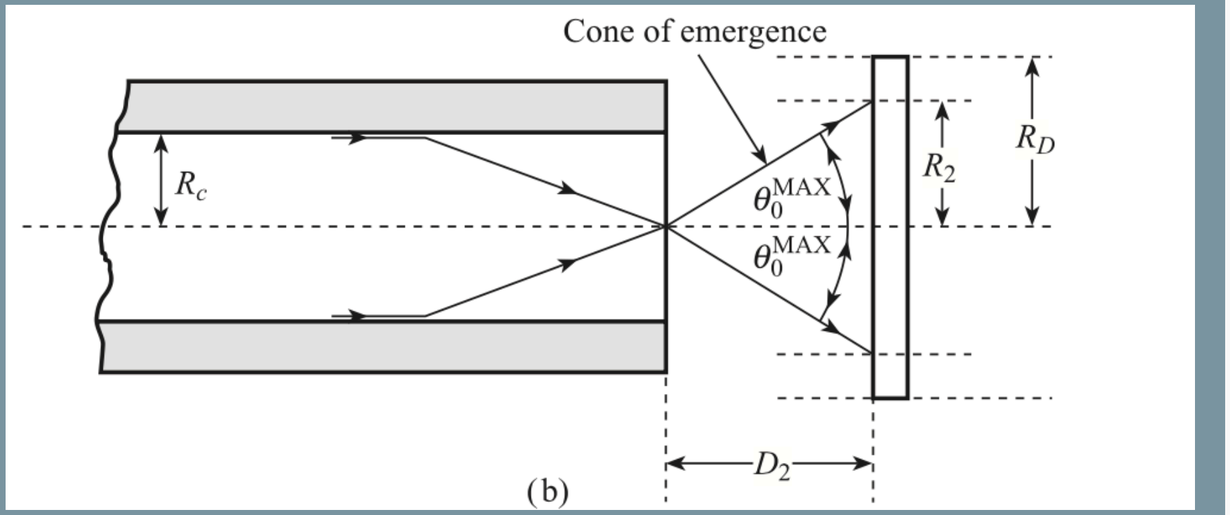
(c) Angular displacement

Here the second fibre is held at a small tilt angle θ relative to the first. Tilting reduces the fraction of rays from the first fibre that fall within the numerical-aperture cone of the second, so the received intensity decreases roughly in proportion to θ (for small angles) as the beams become increasingly misaligned.

(d) Differential arrangement

Two identical “receiver” fibres lie symmetrically about the emitter. A small transverse shift of the emitter towards one side increases coupling into that receiver while decreasing coupling into the other. By taking the difference of the two detector signals, one obtains a signal linearly proportional to displacement but largely immune to source-power fluctuations or common-mode misalignment.

- R_c: Radius of the core of the optical fiber.
- R_2: Radius of the emergence cone at the output face (fiber end).
- R_D: Radius of the detector (receiving the light).
- D_2: Distance from the fiber end to the detector plane.
- θ_0^{MAX}: Maximum acceptance/emergence angle of light from the fiber (half-angle of the cone).
- ω: Power collected by the detector.
- ω_0: Power collected ideally at the reference output plane.
- $W_D(\lambda)$: Spectral density of power at the detector.
- $W_{OM}(\lambda)$: Spectral density of power at the output of the fiber.
- K_{MD}: Mismatch coefficient between the emerging light cone and the detector area.
- NA: Numerical Aperture of the fiber, approximately $\tan(\theta_0^{\mathrm{MAX}})$.

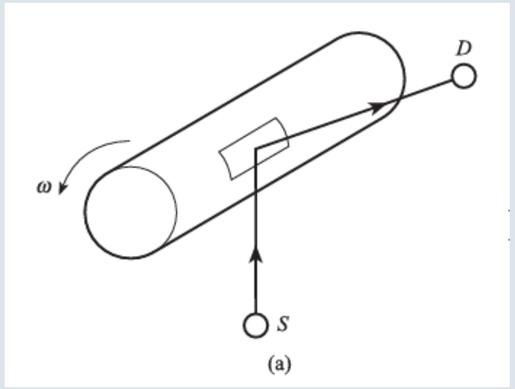


$$W_D(\lambda) = \left(\frac{\omega}{\omega_0}\right) W_{OM}(\lambda), \text{ i.e. } K_{MD} = \left(\frac{\omega}{\omega_0}\right)$$

$$\text{Since } \omega \approx \frac{\pi R_D^2}{D_2^2},\ \omega_0 = \frac{\pi R_2^2}{D_2^2} \text{ and NA} \approx \tan\theta_0^{\mathrm{MAX}},$$

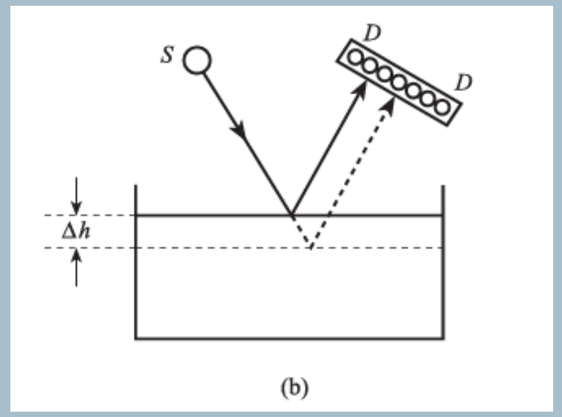
$$K_{MD} \approx \frac{R_D^2}{R_2^2} \approx \frac{R_D^2}{D_2^2(\mathrm{NA})^2}$$

Intensity modulation using reflection



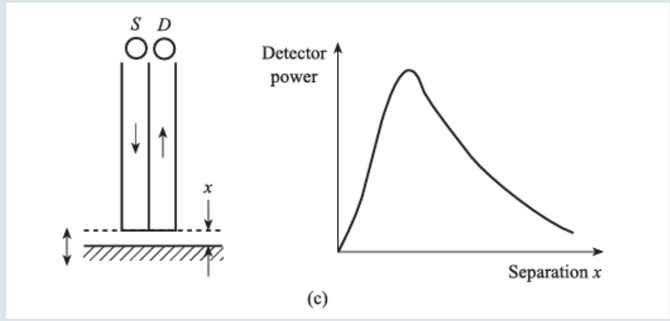
a) Analysis Velocity sensor

- A reflective strip is attached to the rotating mechanism.
- Once per revolution, the strip reflects light back to the detector.
- The number of pulses per second = number of revolutions per second (i.e., angular velocity).
- Simple, remote measurement technique.



b) Level sensor

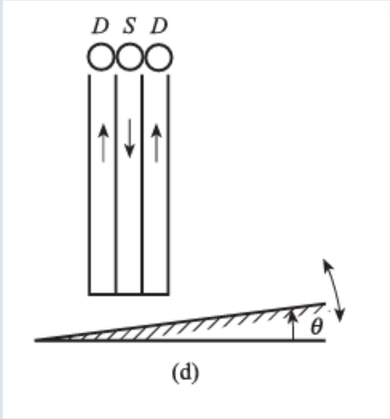
- Beam is reflected from the liquid surface.
- Change in liquid level (Δh) causes lateral displacement of the reflected beam.
- Displacement is proportional to Δh.
- A two-dimensional detector array senses this beam movement.



(c) Two-fibre linear displacement sensor

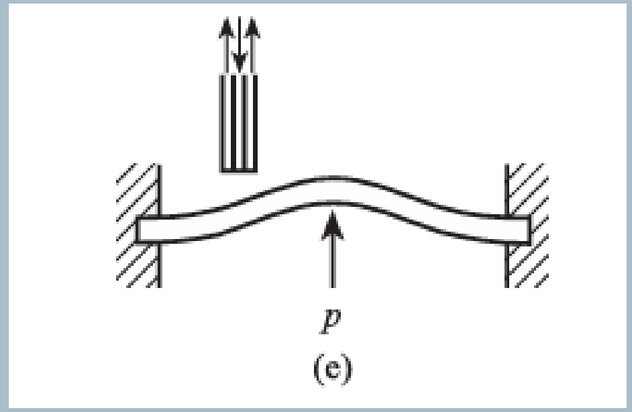
- Two optical fibres: one for sending, one for receiving light.
- Detector power vs separation (x) is non-linear:
- Small x: Power increases as cones of light overlap more.
- Large x: Power decreases following an inverse square law.
- Limitations:.
- Affected by vibration, intensity, reflection.

Intensity modulation using reflection



d) Three-Fibre Angular Displacement Sensor

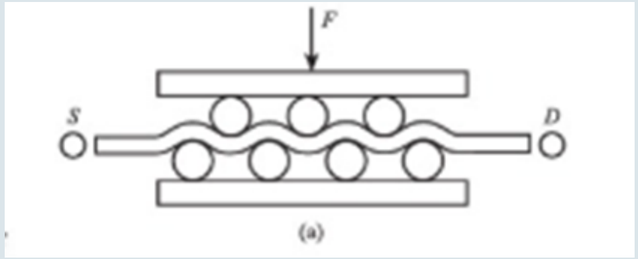
- One source fibre and two detector fibres.
- Angular change (θ) alters the ratio of light collected by two detectors.
- The ratio depends only on θ and fibre refractive indices.
- Independent of source intensity, reflection coefficient, and separation.
- Offers higher accuracy compared to two-fibre systems



e) Three-Fibre Pressure Sensor

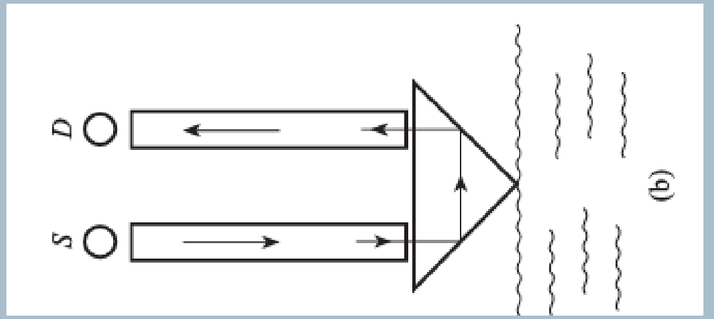
- Based on three-fibre angular displacement sensing.
- Elastic diaphragm bends under pressure.
- Fibre system placed at the diaphragm's maximum angular deformation point.
- Pressure measurement is obtained from changes in the angle sensed by fibres.

Intensity modulation using attenuation and absorption



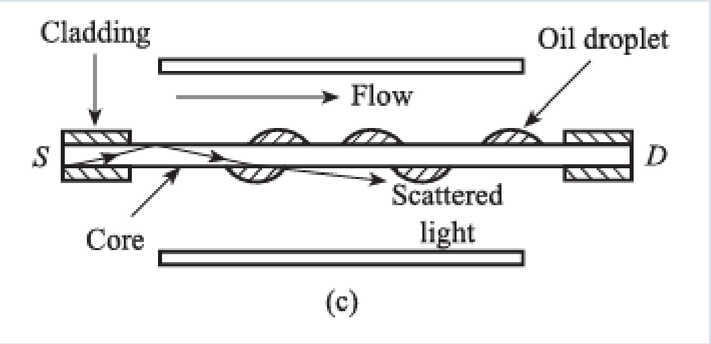
a) Micro bend forcesensor

- Force (F) causes fibre microbending.
- More bending → more light loss into cladding.
- Detector output decreases with force.



b) Two-fibre level detector

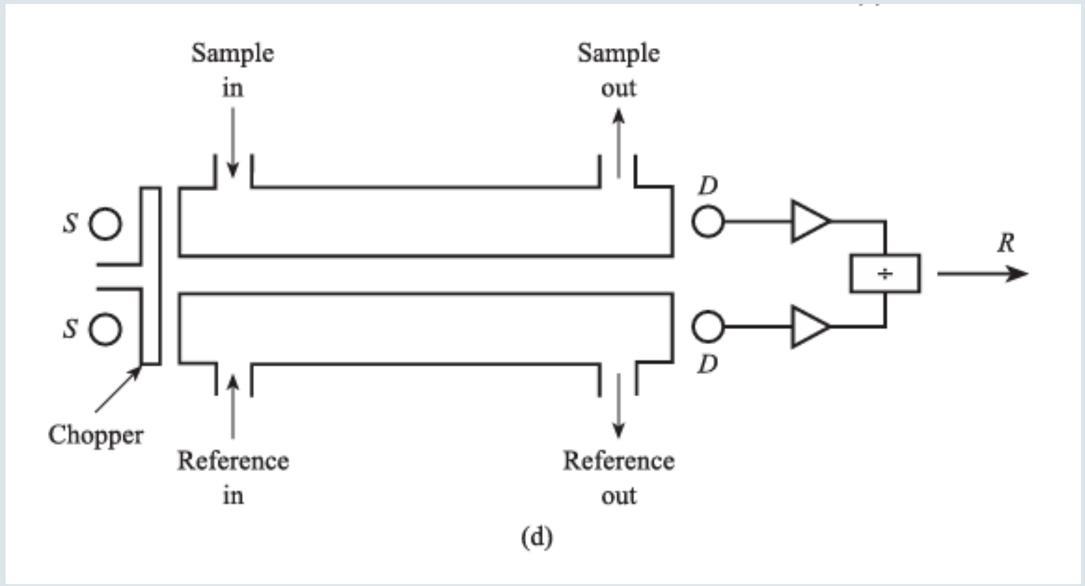
- Two fibres attached to 90° glass prism.
- Low level: Air around prism → total reflection → high signal.
- High level: Liquid around prism → partial reflection → low signal



(c) Oil-in-water detector

- Water flows over fibre core (no cladding).
- Oil droplet on core refracts light out.
- Detector output drops if oil is present.

Intensity modulation using attenuation and absorption



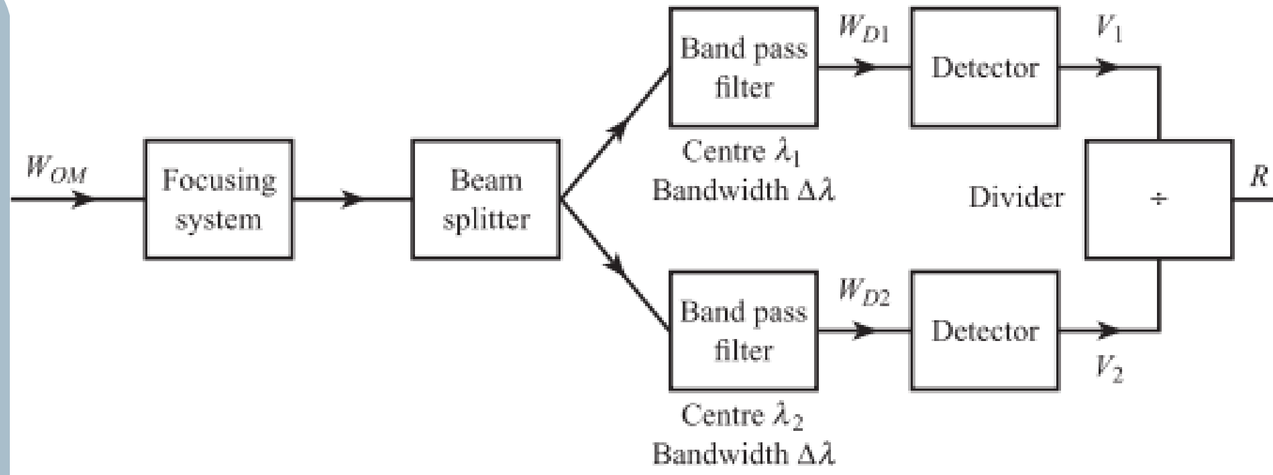
d) Three-Fibre Angular Displacement Sensor

- Detects gases by measuring infrared (IR) absorption.
- Uses two cells:
- Sample cell: contains gas to be measured.
- Reference cell: contains pure reference gas (e.g., CO).
- IR source emits radiation through both cells.
- Absorbing gases reduce the IR transmission.
- Chopper rotates to modulate the light (produces a.c. signals).
- Detectors pick up signals; noise is filtered electronically.
- Ratio (R) of detector signals is calculated.
- R depends on gas concentration, not on source fluctuations.

Two-wavelength systems

In basic intensity-modulated measurement systems (like broadband or narrowband pyrometers), the measurements can get disturbed due to:

- Changes in emissivity of the object (how much radiation it emits)
- Changes in the transmission medium (like dust, gas, or water vapor in between)



SOLUTION :TWO WAVELENGTH SYSTEM

The incoming radiation is focused and split into two beams, each filtered at different center wavelengths (λ_1 and λ_2).

Both beams are detected separately by matched detectors, and an electronic divider calculates the ratio $R = V_1/V_2$.

By using the ratio, errors due to emissivity and transmission changes effectively cancel out, improving measurement accuracy.

$$R(T) = \frac{V_1(T)}{V_2(T)} = \frac{\frac{1}{\pi}(C_1 \varepsilon T_M K_F F K_D)_{\lambda_1} \frac{\Delta\lambda}{\lambda_1^5} \exp\left(-\frac{C_2}{\lambda_1 T}\right)}{\frac{1}{\pi}(C_1 \varepsilon T_M K_F F K_D)_{\lambda_2} \frac{\Delta\lambda}{\lambda_2^5} \exp\left(-\frac{C_2}{\lambda_2 T}\right)}$$

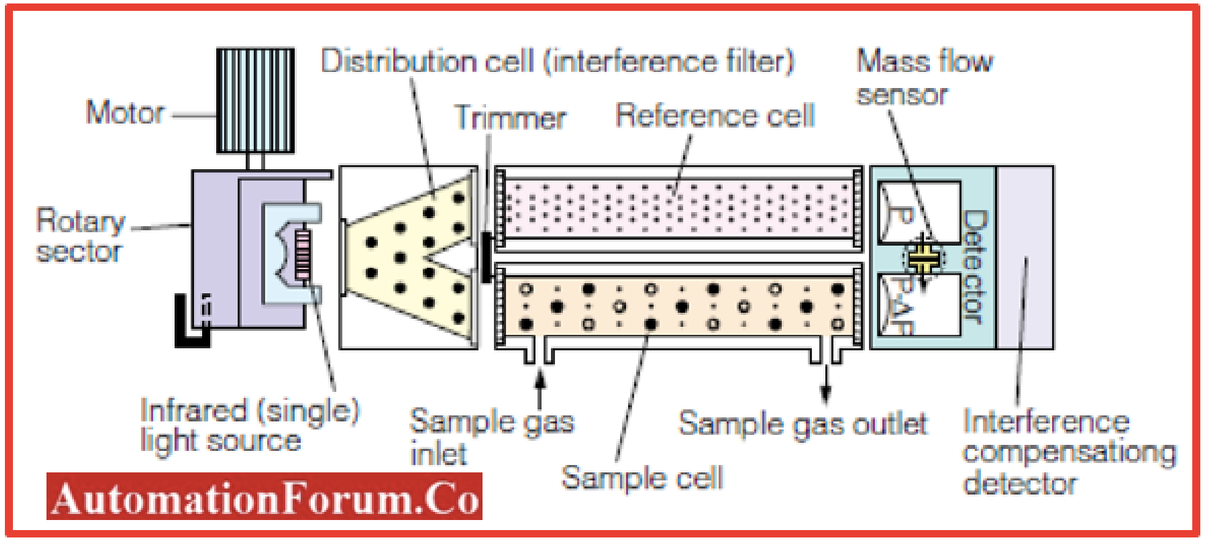
If $\lambda 1$ and $\lambda 2$ are close together, so that the emissivities and transmission factors are equal, then the ratio becomes:

$$R(T) = \left(\frac{\lambda_2}{\lambda_1}\right)^5 \exp \frac{C_2}{T}\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)$$

Advantages and Applications

:

- No need for calibration or emissivity correction.
- Immune to transmission medium fluctuations.
- Higher accuracy under varying conditions.
- Can also be used for infrared gas analysis (measure absorption at different λ for gas components).



Applications:

- Industrial temperature monitoring
- Combustion analysis
- Gas detection in hazardous environments

The two-wavelength principle is also used in IR gas analyzers.

Instead of two beams, we have one beam passing through a gas cell.

λ_1 is chosen to match the absorption peak of the gas of interest (e.g., CO_2, CH_4).

λ_2 is chosen where no absorption occurs (reference point).

By measuring the intensity at these two wavelengths, we can accurately detect gas concentration without being affected by environmental noise.

Wavelength Modulation Systems – Introduction

Why Wavelength Modulation?

Intensity-modulated systems are sensitive to unwanted variations (source intensity, transmission medium).

SOLUTION TO ABOVE

Use wavelength modulation — measured variable directly controls the wavelength of light.

Makes the system independent of source and medium variations.

Advantages:

✔ Immune to source and transmission variations.

✔ High accuracy and stability.

Two key systems developed:

→ Littrow Diffraction Grating (Angular Displacement Sensor)

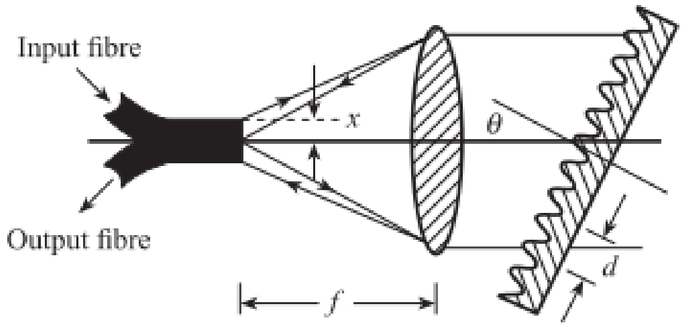
→ Fresnel Zone Plate (Linear Displacement Sensor)

Types of Wavelength Modulation Systems

Littrow Diffraction Grating

Angular Displacement Sensor

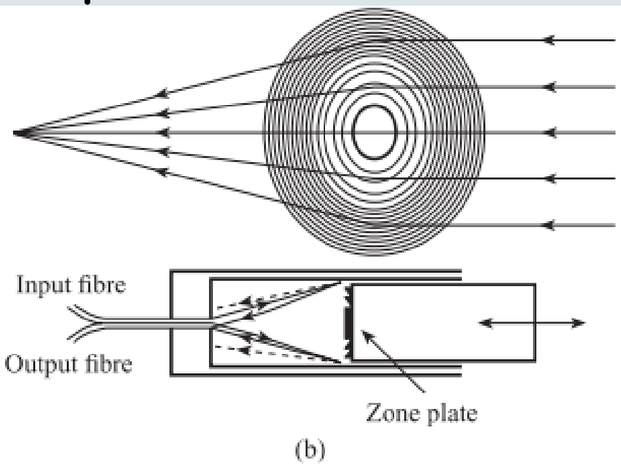
- Broadband light reflected by a grating at angle θ.
- Only specific λ satisfy: 2d sin θ = mλ (usually m=1).
- Wavelength λ varies uniquely with angle θ.
- Used for angular measurement (20°–60° range).



Fresnel Zone Plate – Linear

Displacement Sensor

- Zone plate focuses different wavelengths at different points.
- Focal length f ∝ 1/λ.
- Reflected narrowband wavelength encodes linear displacement x.
- λ related to position: λ = 2d sin θ.



Thank you